24900175

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

Discipline Specific Core Course – MG1DSCMAT100 – GROUND ROOTS OF MATHEMATICS

SCHEME OF VALUATION

Duration: 2 Hours **QP Code: 24900175** **Maximum Marks: 70**

[Learning Domain] [CO No(s)]

Part A

Short Answer Type Questions

Answer Any FIVE Questions. Each question carries 2 Marks

1. $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$ ---------- 1 Mark [K] [1]
   $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$ ---------- 1 Mark
2. Stating any one of the identity laws $(p \wedge T \equiv p;\ \ p \vee F \equiv p)$ 1 Mark [U] [1]
   Proving stated law ---------- 1 Mark
3. $(f \circ g)(x) = f\big(g(x)\big)$ ---------- 1 Mark [U] [2]
   Domain of $f \circ g$: those x in the domain of g for which $g(x)$ lies in the domain of f
   1 Mark
4. $f(-x) = -f(x)$ ---------- 1 Mark [K] [2]
   Example ---------- 1 Mark
5. $\frac{dy}{dx} = 2\sin x \cos x$ ---------- 2 Marks [A] [4]
6. $f'(x) = 2\cos 2x$ ---------- 2 Marks [U] [4]
7. $0 \cdot \infty$ form (It is an indeterminate form) ---------- 2 Marks [A] [6]
8. $f(x_1) < f(x_2)$ whenever $x_1 > x_2$ ---------- 1 Mark [U] [5]
   Example ---------- 1 Mark

Part B

Short Essay Type Questions

Answer Any FIVE Questions. Each question carries 6 Marks

9. a) All rabbits Hop ---------- 3 Marks [U] [1]
   b) All Animals are rabbits and hops ---------- 3 Marks

10. For each definition with example --- 2 marks -------- Total 6 Marks [K] [1]

11. $\lim_{x\to+\infty} \frac{x^2}{e^{3x}} = \lim_{x\to+\infty} \frac{2x}{3e^{3x}}$ ---------- 2 Marks [A] [6]

$= \lim_{x\to+\infty} \frac{2}{9e^{3x}}$ ---------- 2 Marks

$= 0$ ---------- 2 Marks

12. (i) Greatest integer function: The greatest integer less than or equal to x – 2 Marks [U] [2]
Least integer function: The smallest integer greater than or equal to x – 2 Marks
(ii) $\lfloor x \rfloor = \lceil x \rceil$ only when x is an integer. ---------- 2 Marks

13. $\frac{dy}{dx} + \cos y \frac{dy}{dx} = 1$ ---------- 2 Marks [A] [4]
$\frac{dy}{dx} = \frac{1}{1+\cos y}$ ---------- 1 Mark
$\frac{d^2y}{dx^2} = \frac{\sin y}{(1+\cos y)^2}\frac{dy}{dx}$ ---------- 2 Marks
$= \frac{\sin y}{(1+\cos y)^3}$ ---------- 1 Mark

14. $\frac{dr}{ds} = -\frac{2}{3s^3} + \frac{5}{2s^2}$ ---------- 6 Marks [A] [4]

15. $\lim_{x\to 0} \frac{\sin x}{\tan x} = \lim_{x\to 0} \frac{\sin x}{\left(\frac{\sin x}{\cos x}\right)} = \lim_{x\to 0} \cos x = 1$ ---------- 3 Marks [A] [6]
$\lim_{x\to 0} \frac{\sin x}{\tan x} = \lim_{x\to 0} \frac{\cos x}{\sec^2 x} = 1$ ---------- 3 Marks

16. $\lim_{x\to+\infty} \frac{x^2}{e^{3x}} = \lim_{x\to+\infty} \frac{2x}{3e^{3x}}$ ---------- 2 Marks [A] [6]
$= \lim_{x\to+\infty} \frac{2}{9e^{3x}}$ ---------- 2 Marks
$= 0$ ---------- 2 Marks

Part C
Essay Type Questions
Answer Any <u>THREE</u> Questions. Each question carries 10 Marks

17. 2 marks for each correct answer ---------- Total 10 Marks [U] [1]

18. (a) (i) $f \circ g = x^2 + 2$ ---------- 2 Marks [U] [2]
(ii) $g \circ f = (x+5)^2 - 3$ ---------- 2 Marks
(iii) $f \circ f = x + 10$ ---------- 1 Mark
(iv) $g \circ g = (x^2-3)^2 - 3$ ---------- 1 Mark
(b) (i) $y = \sqrt{\frac{x}{3}}$ or for any reasonable attempt --------- 2 Marks
(ii) $y = \frac{1}{3}\sqrt{x}$ or $3y = \sqrt{x}$ ---------- 2 Marks

19. (a) $f^{-1}(x) = \frac{x-5}{2}$ ---------- 4 Marks [An] [3]
(b) Getting $f\left(f^{-1}(x)\right) = x$ and $f^{-1}\left(f(x)\right) = x$ ----- 3 Marks
(c) Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$ ---------- 3 Marks

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20. (a) $y' = \frac{2x^3+17x^2+20x+1}{(x+5)^2}$ ---------- 5 Marks [A] [4]

(b) $y' = \frac{-2x^5+4x^3+2x}{(x^4+1)^2}$ ---------- 5 Marks

21. $f'(x) = 2\cos 2x$ ---------- 1 Mark [U] [5]

Critical points are $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ ---------- 2 Marks

$f"(x) = -4\sin 2x$ ---------- 1 Mark

$f"(\frac{\pi}{4}) = -4 < 0$, Maximum exists at $x = \frac{\pi}{4}$ ------- 2 Marks

The maximum value is $f\left(\frac{\pi}{4}\right) = 1$ ---------- 1 Mark

$f"(\frac{3\pi}{4}) = 4 > 0$, Minimum exists at $x = \frac{3\pi}{4}$ -------- 2 Marks

The minimum value is $f\left(\frac{3\pi}{4}\right) = -1$ ---------- 1 Mark

22. $f''(x) = 6x - 6$ ---------- 2 Marks [U] [5]

$f''(x) = 0 \Rightarrow x = 1$ ---------- 2 Marks

(a) concave up on $(1, \infty)$ ---------- 2 Marks

(b) concave down on $(-\infty, 1)$ ---------- 2 Marks

(c) inflection point $(1, -99)$ ---------- 2 Marks

NB: Stage marks for any correct alternate method can be fixed by the examiner.